MODEL SUBIECT EXAMEN PARTIAL
MATE 2

① Fie $A = \begin{pmatrix} 1 & 0 & 1 \\ 3 & 1 & 0 \\ -1 & -1 & 2 \end{pmatrix}$

a) Det. B_1 bază în $W_1 = \{ v \in \mathbb{R}^3 \mid A^t \cdot v = 0 \}$

b) Det. B_2 bază în W_2 = sp. generat de col. matr. A.

Arătați că mult. $B = B_1 \cup B_2$ este bază în $\mathbb{R}^3$ și det. coord. vect. $v = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$ în baza B $([v]_B)$.

② Fie aplic. liniară $T: \mathbb{R}^3 \to \mathbb{R}^3$ dată de:

$$T\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x - 2y + z \\ -2x + 4y - 2z \\ -x + 2y - z \end{pmatrix}$$

Determinați $[T]_{B_c}$ (baza canonică) și arătați că $\nexists\, \alpha \in \mathbb{R}$ a.î.

$$T \circ T = \alpha T.$$

Determinați $\ker T$, $\operatorname{Im} T$ (bază, dimens.)

③ Se consideră matricea:

$$A = \begin{pmatrix} 0,4 & 0,7 \\ 0,6 & 0,3 \end{pmatrix}$$

a) Arătați că A este diagonalizabilă; Aduceți la forma diag.

b) Folosiți rez. anterior și arătați că:

$\lim_{n \to \infty} A^n$ este o matr. cu coloane egale

④ Fie W mult. soluțiilor sist:

$$\begin{cases} x - 3y + 2z = 0 \\ 2x + y - 3z = 0 \\ -3x + 2y + z = 0 \end{cases}$$

a) Det. o bază ortogonală în W și prelungiți-o la o bază ortogonală în $\mathbb{R}^3$. (notată B).

b) Fie $v = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$. Determinați $pr_W v$ și $[v]_B$.

Rezolvări

① a) $A^t = \begin{pmatrix} 1 & 3 & -1 \\ 0 & 1 & -1 \\ 1 & 0 & 2 \end{pmatrix}$ Fie $v = \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3$.

$A^t \cdot v = 0 \iff \begin{pmatrix} 1 & 3 & -1 \\ 0 & 1 & -1 \\ 1 & 0 & 2 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \iff$

$\iff \begin{cases} x + 3y - z = 0 \\ y - z = 0 \\ x + 2z = 0 \end{cases} \Rightarrow \boxed{y = z} \Rightarrow \begin{cases} x + 2y = 0 \\ x + 2y = 0 \end{cases}$ Notez $y = \alpha \Rightarrow z = \alpha$, $x = -2\alpha$

$\Rightarrow v = \begin{pmatrix} -2\alpha \\ \alpha \\ \alpha \end{pmatrix} = \alpha \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix}$.

$W_1 = \mathcal{L}\left\{ \alpha \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix}, \alpha \in \mathbb{R} \right\} \Rightarrow \dim W_1 = 1$ și $B_1 = \underline{\underline{\begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix}}}$.

b) B_2 = bază în sp. gen. de col. lui A

$W_2 = \mathcal{L}\left\{ \underbrace{\begin{pmatrix} 1 \\ 3 \\ -1 \end{pmatrix}}_{u_1}, \underbrace{\begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}}_{u_2}, \underbrace{\begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}}_{u_3} \right\}$ $\{u_1, u_2, u_3\}$ sist. generatori

$\{u_1, u_2, u_3\}$ l.i $\iff \operatorname{rang} A = 3$

$\det A = \begin{vmatrix} 1 & 0 & 1 \\ 3 & 1 & 0 \\ -1 & -1 & 2 \end{vmatrix} = 2 - 3 + 1 = 0 \Rightarrow \operatorname{rang} A \leq 2$ ($\operatorname{rang} A = 2$, $\exists$ minor ord 2 nenul).

Aleg 2 vect. l.i: $\{u_1, u_2\}$ $\alpha_1 u_1 + \alpha_2 u_2 = 0 \iff \alpha_1 = \alpha_2 = 0$

$\begin{cases} \alpha_1 = 0 \\ 3\alpha_1 + \alpha_2 = 0 \\ -\alpha_1 - \alpha_2 = 0 \end{cases} \Rightarrow \alpha_1 = \alpha_2 = 0.$

$\Rightarrow B_2 = \left\{ \underbrace{\begin{pmatrix} 1 \\ 3 \\ -1 \end{pmatrix}}_{u_1}, \underbrace{\begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}}_{u_2} \right\}$

• $B = B_1 \cup B_2 = \left\{ \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 3 \\ -1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} \right\}$ sist. generatori; dacă sunt l.i $\Rightarrow$ bază.

Fie $B = \begin{pmatrix} -2 & 1 & 0 \\ 1 & 3 & 1 \\ 1 & -1 & -1 \end{pmatrix}$ $\operatorname{rang} B = 3$ ($\det B \neq 0$)

$\Rightarrow B$ bază în $\mathbb{R}^3$.

Notez coord. lui $v = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$ în baza B cu x, y, z.

$\Rightarrow x \cdot \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix} + y \begin{pmatrix} 1 \\ 3 \\ -1 \end{pmatrix} + z \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$ $\rightarrow$

$$\Rightarrow \begin{cases} -2x+y=0 \\ x+3y+z=1 \\ x-y-z=0 \end{cases} \rightarrow y=2x \sim \begin{cases} 7x+z=1 \\ -x-z=0 \end{cases} \oplus$$

$$6x = 1 \Rightarrow x=\frac{1}{6} \rightarrow y=\frac{1}{3} \Rightarrow z=-\frac{1}{6}$$

$$\Rightarrow [v]_B = \begin{pmatrix} 1/6 \\ 1/3 \\ -1/6 \end{pmatrix}.$$

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(2) $T\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x-2y+z \\ -2x+4y-2z \\ -x+2y-z \end{pmatrix}$.

Fie A = matr. asoc transf liniare T: $A = \begin{pmatrix} 1 & -2 & 1 \\ -2 & 4 & -2 \\ -1 & 2 & -1 \end{pmatrix}$.

$$T\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 & -2 & 1 \\ -2 & 4 & -2 \\ -1 & 2 & -1 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix}. \quad [T]_{Bc} = A, \text{ deoarece:}$$

$$T(e_1) = A\cdot\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ -2 \\ -1 \end{pmatrix}$$

$$T(e_2) = A\cdot\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -2 \\ 4 \\ 2 \end{pmatrix}$$

$$T(e_3) = A\cdot\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ -2 \\ -1 \end{pmatrix}.$$

$$[T]_{Bc} = (T(e_1), T(e_2), T(e_3)) = A$$

$T\circ T = \alpha T$; Fie $v \in \mathbb{R}^3$.

$$T\circ T(v) = T(T(v)) = T(A\cdot v) = A\cdot(A\cdot v) = A^2\cdot v.$$

$$\alpha T(v) = \alpha\cdot A\cdot v$$

$$\Rightarrow T\circ T = \alpha T \Leftrightarrow \alpha A = A^2.$$

$$A^2 = A\cdot A = \begin{pmatrix} 1 & -2 & 1 \\ -2 & 4 & -2 \\ -1 & 2 & -1 \end{pmatrix}\begin{pmatrix} 1 & -2 & 1 \\ -2 & 4 & -2 \\ -1 & 2 & -1 \end{pmatrix} = \begin{pmatrix} 4 & -8 & 4 \\ -8 & 16 & -8 \\ -4 & 8 & -4 \end{pmatrix} =$$

$$= 4\cdot A.$$

$$\Rightarrow \alpha A = 4A \Rightarrow \boxed{\alpha = 4}.$$

$$\operatorname{Ker} T = \{ v \in \mathbb{R}^3 \mid T(v) = 0 \}.$$

$T(v) = 0 \Leftrightarrow A\cdot v = 0$, unde $v = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$.

$$\Rightarrow \begin{cases} x-2y+z=0 \\ -2x+4y-2z=0 \\ -x+2y-z=0 \end{cases}$$

Rez. sist. cu Gauss:

$$\left(\begin{array}{ccc|c} \boxed{1} & -2 & 1 & 0 \\ -2 & 4 & -2 & 0 \\ -1 & 2 & -1 & 0 \end{array}\right) \begin{matrix} \\ L_2+2L_1 \\ L_3+L_1 \end{matrix} \left(\begin{array}{ccc|c} 1 & -2 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right) \quad -3-$$

$\Rightarrow x - 2y + z = 0$

Notez $z = \alpha$, $y = \beta$

$\Rightarrow x = 2\beta - \alpha$.

$$\text{Ker } T = \mathcal{L}\left\{ \begin{pmatrix} 2\beta-\alpha \\ \beta \\ \alpha \end{pmatrix}, \alpha, \beta \in \mathbb{R} \right\} =$$

$$= \mathcal{L}\left\{ \alpha \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} + \beta \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}, \alpha, \beta \in \mathbb{R} \right\}.$$

$\Rightarrow \dim(\text{Ker } T) = 2$

o bază în $\text{Ker } T$: $\mathcal{B}_{\text{Ker } T} = \left\{ \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} \right\}$.

Din T. Rang-Defect:

$\dim(\text{Ker } T) + \dim(\text{Im } T) = \dim(\mathbb{R}^3)$

$\Rightarrow \dim(\text{Im } T) = 1$.

Deoarece $\text{Im } T$ este sp. gen. de col. matr. A și $\dim(\text{Im } T) = 1$

$\Rightarrow$ aleg pt. bază lui $\text{Im } T$, o col. din A.

de ex: $\mathcal{B}_{\text{Im } T} = \left\{ \begin{pmatrix} 1 \\ -2 \\ -1 \end{pmatrix} \right\}$.

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(3) $A = \begin{pmatrix} 0,4 & 0,7 \\ 0,6 & 0,3 \end{pmatrix}$.

Polinom caracteristic: $P_A(\lambda) = \det(A - \lambda I_2) = \begin{vmatrix} 0,4-\lambda & 0,7 \\ 0,6 & 0,3-\lambda \end{vmatrix} =$

$= (0,4-\lambda)(0,3-\lambda) - 0,42 = \lambda^2 - 0,7\lambda - 0,3$.

$P_A(\lambda) = 0 \Rightarrow \lambda_1 = 1 \quad m_a(\lambda_1) = 1$ (multipl. algebrice).

$\lambda_2 = -0,3. \quad m_a(\lambda_2) = 1$

Det. vect. proprii:

- $\boxed{\lambda_1 = 1}$ $V_{\lambda_1} = \{ v \in \mathbb{R}^2 \mid A \cdot v = \lambda_1 v \} = \{ v \in \mathbb{R}^2 \mid (A - \lambda_1 I_2) v = 0 \}$

Fie $v = \begin{pmatrix} x \\ y \end{pmatrix} \Rightarrow \begin{cases} -0,6x + 0,7y = 0 \\ 0,6x - 0,7y = 0 \end{cases}$

$A - \lambda_1 I_2 = \begin{pmatrix} -0,6 & 0,7 \\ 0,6 & -0,7 \end{pmatrix}$

$\Rightarrow$ Notez $y = \alpha$

$x = \frac{7}{6}\alpha$.

$\Rightarrow V_{\lambda_1} = \mathcal{L}\left\{ \alpha \begin{pmatrix} 7/6 \\ 1 \end{pmatrix}, \alpha \in \mathbb{R} \right\}$. $\dim V_{\lambda_1} = 1$

$\Rightarrow m_g(\lambda_1) = 1$.

• $\boxed{\lambda_2 = -0{,}3}$ $\quad V_{\lambda_2} = \{ v \in \mathbb{R}^2 / Av = \lambda_2 v \} = \{ v \in \mathbb{R}^2 / (A - \lambda_2 I_2) v = 0 \}$

Fie $v = \begin{pmatrix} x \\ y \end{pmatrix}$ $\Rightarrow \begin{cases} 0{,}7x + 0{,}7y = 0 \\ 0{,}6x + 0{,}6y = 0 \end{cases}$ $\Rightarrow$ Notez $y = \alpha$ $\Rightarrow x = -\alpha$.

$A - \lambda_2 I_2 = \begin{pmatrix} 0{,}7 & 0{,}7 \\ 0{,}6 & 0{,}6 \end{pmatrix}$

$\Rightarrow V_{\lambda_2} = \mathcal{L} \left\{ \alpha \begin{pmatrix} -1 \\ 1 \end{pmatrix}, \alpha \in \mathbb{R} \right\}$ $\quad \dim V_{\lambda_2} = 1 \Rightarrow m_g(\lambda_2) = 1$.

$\Rightarrow A$ diagonalizabilă deoarece $m_a(\lambda_1) = m_g(\lambda_1) = 1$
$m_a(\lambda_2) = m_g(\lambda_2) = 1$.

Fie $D = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -0{,}3 \end{pmatrix}$.

$A = T \cdot D \cdot T^{-1}$ unde $T = (V_{\lambda_1}, V_{\lambda_2}) = \begin{pmatrix} 7 & -1 \\ 6 & 1 \end{pmatrix}$

[iau $\alpha = 6 \Rightarrow \begin{pmatrix} 7 \\ 6 \end{pmatrix}$].

Calculez T^{-1} (Gauss-Jordan):

$$\left(\begin{array}{cc|cc} 7 & -1 & 1 & 0 \\ 6 & 1 & 0 & 1 \end{array}\right) \xrightarrow{L_2 - \frac{6}{7}L_1} \left(\begin{array}{cc|cc} 7 & -1 & 1 & 0 \\ 0 & \frac{13}{7} & -\frac{6}{7} & 1 \end{array}\right) \xrightarrow{\frac{7}{13}L_2}$$

$$= \left(\begin{array}{cc|cc} 7 & -1 & 1 & 0 \\ 0 & 1 & -\frac{6}{13} & \frac{7}{13} \end{array}\right) \xrightarrow{L_1 + L_2} \left(\begin{array}{cc|cc} 7 & 0 & \frac{7}{13} & \frac{7}{13} \\ 0 & 1 & -\frac{6}{13} & \frac{7}{13} \end{array}\right) \xrightarrow{\frac{1}{7}L_1}$$

$$= \left(\begin{array}{cc|cc} 1 & 0 & \frac{1}{13} & \frac{1}{13} \\ 0 & 1 & -\frac{6}{13} & \frac{7}{13} \end{array}\right) \Rightarrow \quad T^{-1} = \frac{1}{13} \begin{pmatrix} 1 & 1 \\ -6 & 7 \end{pmatrix}.$$

$$A = T \cdot D \cdot T^{-1} = \frac{1}{13} \cdot \begin{pmatrix} 7 & -1 \\ 6 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -0{,}3 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ -6 & 7 \end{pmatrix} =$$

$$= \frac{1}{13} \begin{pmatrix} 7 & 0{,}3 \\ 6 & -0{,}3 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ -6 & 7 \end{pmatrix} = \frac{1}{13} \begin{pmatrix} 5{,}2 & 9{,}1 \\ 7{,}8 & 3{,}9 \end{pmatrix}.$$

$A^n = T \cdot D^n \cdot T^{-1}$ $\quad D^n = \begin{pmatrix} 1 & 0 \\ 0 & (-0{,}3)^n \end{pmatrix}$.

$$A^n = \frac{1}{13} \begin{pmatrix} 7 & -1 \\ 6 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & \left(-\frac{3}{10}\right)^n \end{pmatrix} \begin{pmatrix} 1 & 1 \\ -6 & 7 \end{pmatrix} = \frac{1}{13} \begin{pmatrix} 7 & -\left(-\frac{3}{10}\right)^n \\ 6 & \left(-\frac{3}{10}\right)^n \end{pmatrix} \begin{pmatrix} 1 & 1 \\ -6 & 7 \end{pmatrix},$$

$$= \frac{1}{13} \begin{pmatrix} 7 + 6 \cdot \left(-\frac{3}{10}\right)^n & 7 - 7 \cdot \left(-\frac{3}{10}\right)^n \\ 6 - 6 \cdot \left(-\frac{3}{10}\right)^n & 6 + 7 \cdot \left(-\frac{3}{10}\right)^n \end{pmatrix} \xrightarrow{n \to \infty} \frac{1}{13} \begin{pmatrix} 7 & 7 \\ 6 & 6 \end{pmatrix}.$$

(4) $\begin{cases} x - 3y + 2z = 0 \\ 2x + y - 3z = 0 \\ -3x + 2y + z = 0. \end{cases}$ W = mult. soluțiilor.

Rez. sist. cu met. Gauss:

$$\left(\begin{array}{ccc|c} \boxed{1} & -3 & 2 & 0 \\ 2 & 1 & -3 & 0 \\ -3 & 2 & 1 & 0 \end{array}\right) \begin{matrix} L_2 - 2L_1 \\ L_3 + 3L_1 \end{matrix} \left(\begin{array}{ccc|c} 1 & -3 & 2 & 0 \\ 0 & 7 & -7 & 0 \\ 0 & -7 & 7 & 0 \end{array}\right) \Rightarrow$$

$$\begin{cases} -7y + 7z = 0 \Rightarrow \boxed{y = z} \\ 7y - 7z = 0 \\ x - 3y + 2z = 0 \Rightarrow x - y = 0 \quad \boxed{x = y} \end{cases}$$

Notez $z = \alpha$
$\Rightarrow x = y = z = \alpha$

$$W = \mathcal{L}\left\{\alpha \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \alpha \in \mathbb{R}\right\}. \quad \dim W = 1.$$

a) Bază ortogonală în W: $B_W = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ ($\dim W = 1 \Rightarrow$ orice bază e ortogonală)

Prelungire la o bază ortogonală în $\mathbb{R}^3$:

1) Mai întâi prelung. la o <u>bază</u> în $\mathbb{R}^3$, folosind vect. bazei canonice

$$B = \left\{\underbrace{\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}}_{v_1}, \underbrace{\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}}_{v_2}, \underbrace{\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}}_{v_3}\right\} \text{ refl. } \{v_1, v_2, v_3\} \text{ l.i.:}$$

fie $C = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$ $\det C = 1 \Rightarrow \operatorname{rang} C = 3 \Rightarrow$ <u>l.i.</u>

2) Ortogonalizare:

$$u_1 = v_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

$$u_2 = v_2 - pr_{u_1}(v_2) = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} - \frac{v_2 \cdot u_1}{u_1 \cdot u_1} \cdot u_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} - \frac{1}{3}\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} =$$

$$= \begin{pmatrix} 2/3 \\ -1/3 \\ -1/3 \end{pmatrix}$$

$$u_3 = v_3 - pr_{u_1}(v_3) - pr_{u_2}(v_3) = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} - \frac{v_3 \cdot u_1}{u_1 \cdot u_1} \cdot u_1 - \frac{v_3 \cdot u_2}{u_2 \cdot u_2} \cdot u_2 =$$

$$= \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} - \frac{1}{3}\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + \frac{\frac{1}{3}}{\frac{4}{9} + \frac{1}{9} + \frac{1}{9}} \cdot \begin{pmatrix} 2/3 \\ -1/3 \\ -1/3 \end{pmatrix} =$$

$$= \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} - \begin{pmatrix} 1/3 \\ 1/3 \\ 1/3 \end{pmatrix} + \begin{pmatrix} 1/3 \\ -1/6 \\ -1/6 \end{pmatrix} = \begin{pmatrix} 0 \\ 1/2 \\ -1/2 \end{pmatrix} \Rightarrow$$

$\Rightarrow \mathcal{B}_{ortog.} = \left\{ \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 2/3 \\ -1/3 \\ -1/3 \end{pmatrix}, \begin{pmatrix} 0 \\ 1/2 \\ -1/2 \end{pmatrix} \right\}$.

b) $v = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$.

- $pr_W(v) = pr_{u_1}(v) = \frac{v \cdot u_1}{u_1 \cdot u_1} \cdot u_1 = \frac{1}{3} \cdot \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1/3 \\ 1/3 \\ 1/3 \end{pmatrix}$.
- $[v]_{B_{ortog}} = \alpha u_1 + \beta u_2 + \gamma u_3$ (coord. sunt α, β, γ)

$$\Rightarrow \left(\begin{array}{ccc|c} 1 & 2/3 & 0 & 1 \\ 1 & -1/3 & 1/2 & 0 \\ 1 & -1/3 & -1/2 & 0 \end{array}\right) \begin{matrix} \\ L_2 - L_1 \\ L_3 - L_1 \end{matrix} \left(\begin{array}{ccc|c} 1 & 2/3 & 0 & 1 \\ 0 & \boxed{-1} & 1/2 & -1 \\ 0 & -1 & -1/2 & -1 \end{array}\right) \begin{matrix} \\ \\ L_3 - L_2 \end{matrix}$$

$$= \left(\begin{array}{ccc|c} 1 & 2/3 & 0 & 1 \\ 0 & -1 & 1/2 & -1 \\ 0 & 0 & -1/4 & 0 \end{array}\right) \Rightarrow \begin{cases} -\frac{1}{4}\gamma = 0 & \Rightarrow \gamma = 0 \\ -\beta + \frac{1}{2}\gamma = -1 & \Rightarrow \beta = 1 \\ \alpha + \frac{2}{3}\beta = 1 & \alpha = \frac{1}{3}. \end{cases}$$

$$\Rightarrow [v]_{B_{ortog}} = \begin{pmatrix} 1/3 \\ 1 \\ 0 \end{pmatrix}.$$